EL NIÑO ACTION BRIEFING · 30 OCTOBER 2026

Kapa‘au and North Kohala, Hawai‘i Island

In North Kohala, El Niño's danger is a dry winter that cures the grass, then one windy day that brings fire, a power cut, and failing wells together. The first job is knowing, by name, who cannot manage without power or a car.

IMPACT REGION	EXCESS HEAT DEATHS	WINDOW	DOMINANT HAZARD
A Hawai ʻ i County region (inferred; not resolved)	None published for the US, Hawai ʻ i, or Hawai ʻ i County	Global 6-month figure 239,000 ± 11,000 (Sep 2026 – Feb 2027); not scaled here	Winter drought, then grass fire, power and water outages, and smoke

13/13 recommendations checked. 4 confirmed, 9 corrected, 0 unverified, 0 contraindicated. Five practices are flagged do not do. All nine For you steps were checked on the same tests, judged for one household. Exposure figure: Climate Impact Lab, September 2026, pre-peer-review, heat only. No figure is published for anywhere in the US, so none is borrowed or scaled. The funding target fell 30% in verification, from about $206,000 to $144,270, mainly because a $100,000 fuel grant through the wrong channel became a $25,000 match for the state's own wildfire grant.

EXECUTIVE SUMMARY — AT A GLANCE



20.23°N 155.80°W, United States of America · Natural Earth

- **The risk.** Fire closed Route 270 for a day in August 2023, and since 2025 broken wells and power cuts have put Hāwī and Kapa ʻ au on essential-use water three times.
- **For you.** As a resident: by 15 November, sign up for county Everbridge alerts and know both roads out, Route 270 and Route 250; it costs nothing.
- **The money.** $94,270 / $144,270 / $475,620; the best dollar buys multi-day backup power for the 44 people on powered medical equipment.
- **Do not.** Do not wait for a siren or a phone alert: leave at the first advisory or when fire is reported near your road.

For you, as a resident	9 steps	p. 2
What to do	**8** actions	p. 6
Who does what	RACI chart	p. 13
Where money does the most	funding	p. 13
Do not do this	**5** items	p. 15
What has happened here	the record	p. 16
How this was produced	methodology	p. 19
References	sources	p. 20

* Suggested formulation: check with a health worker before sharing.

WHERE THIS IS

SCOPE

North Kohala is the northern tip of Hawai ʻ i Island. Its towns, **Kapa ʻ au** (ZIP 96755) and **Hāwī** (ZIP 96719), sit on the wetter windward side; the subdivisions down the leeward coast (Kohala Ranch, Kohala by the Sea, Kohala Estates, Kohala Waterfront) sit in dry grassland. The request gave a street address in Kapa ʻ au. A single address has no data of its own, so the briefing covers the district, both ZIP codes. Only two roads leave it: **Akoni Pule Highway (Route 270)** south to Kawaihae, and **Kohala Mountain Road (Route 250)** to Waimea. Tap water comes from a few county wells around Hāwī on a dead-end line; many homes outside the towns drink from rain catchment.

The season ahead: NOAA's Climate Prediction Center (8 October) gives better than 83% odds of a strong to very strong El Niño through January–March 2027, and its Hawai ʻ i outlook (15 October) favours below-normal rain at Hilo, 75% odds, for December to February. El Niño winters in Hawai ʻ i tend to be dry from November to April. As of 27 October the US Drought Monitor shows **no drought in Hawai ʻ i**; the dryness is still to come. Grass that grew in this year's rains, and trees Hurricane Lala brought down in August, become the fuel. So **spring and summer 2027 are part of this event**, not only the winter.

FOR YOU, AS A RESIDENT

9 STEPS

For a household in Kapa ʻ au or Hāwī, an El Niño winter means a dry season that can end in fire on the leeward side, and power cuts that stop the county wells, the fridge, and anyone's oxygen at the same time. The ranked actions below are written for the county, the state, and their partners. These nine are the ones you can take yourself, at your own cost, grouped by what they protect. No role was given with the request, so they are written for a resident.

1 · Leaving: sign up for county alerts, know both roads out, and leave at the first advisory, by 15 November

PARTIALLY CONFIRMED FROM ACTIONS 3 AND 4

BY WHEN	COST TO YOU	PROTECTS
15 Nov 2026	$0; about $25–60 for a weather radio (estimate)	Everyone in the house

Hawai ʻ i County Civil Defense sends alerts through Everbridge: sign up at member.everbridge.net/371914103062563/new (linked from hawaiipolice.gov) or call (808) 935-0031. Hawai ʻ i County has no pre-set evacuation zones; orders name the subdivision. In August 2023 fire closed Route 270 between mile markers 6 and 17, and upper Kohala Ranch got out by Kohala Mountain Road. Route 250 itself

* Suggested formulation: check with a health worker before sharing.

has closed for flooding and for Hurricane Lala, so know both, and leave before either closes. The NOAA Weather Radio station for the area is WZ2540, Kamuela, on 162.500 MHz.

2 · Carbon monoxide: fit a battery alarm on every level and outside sleeping areas, and run any generator at least 20 feet outside

CONFIRMED FROM DO NOT, CARD 3

BY WHEN	COST TO YOU	PROTECTS
30 Nov 2026	About $20–50 per alarm (estimate)	Everyone asleep in the house

CDC: never run a generator in a home, garage, or carport, and keep it at least 20 feet from doors, windows, and vents. A lanai counts as too close. During a 2019 California power shutoff, eight people in Campo went to hospital after a generator was placed too near their home.*

3 · Power: check your address on Hawaiian Electric's shutoff map, and if anyone uses oxygen, CPAP, or home dialysis, file the medical-needs form and plan power for several days

PARTIALLY CONFIRMED FROM ACTION 1

BY WHEN	COST TO YOU	PROTECTS
30 Nov 2026	$0 to register; about $1,300 for a battery with a solar panel (estimate)	Oxygen, medical devices, phones

Hawaiian Electric's shutoff hotline is 1-844-483-8666. Its "Medical Needs Communication Form" (pages.hawaiianelectric.com/medical-needs-form) gets you advance notice; it does not exempt you from a shutoff. Kapa ʻau and Hāwī are probably outside the shutoff zone, but most outages here come from wind and falling trees: about 2,100 North Kohala customers lost power in November 2025, and Lala cut power for days in August 2026. A small battery runs an oxygen concentrator for only a few hours, so ask your oxygen supplier for backup cylinders too.*

4 · Drinking water: store 14 days, one gallon per person per day, plus water for pets and animals

* Suggested formulation: check with a health worker before sharing.

CONFIRMED FROM ACTION 2

BY WHEN	COST TO YOU	PROTECTS
15 Dec 2026	About $30–60 for containers (estimate)	Drinking water

That is the Hawai ʻ i Emergency Management Agency's advice, repeated by the county Department of Water Supply in March 2026. North Kohala's taps have been cut to essential use three times since August 2025, each time because a well broke or the power went off, not because the aquifer ran dry.

5 · Catchment: clean and test your tank while it is low, and disconnect the downspout after ash or smoke until rain has washed the roof

PARTIALLY CONFIRMED ROLE-SPECIFIC

BY WHEN	COST TO YOU	PROTECTS
Before the tank runs low	From $25 plus shipping for a subsidised lead and copper test	Water for households not on county lines

The state health department recommends testing catchment water for E. coli, turbidity, lead, and copper, and subsidises an annual lead and copper test through AECOS. The University of Hawai ʻ i's catchment guidance (written for volcanic ash) says to disconnect from the roof until rain flushes it and to clean the gutters. No state list of approved haulers for home tanks exists, so ask a hauler for a food-grade tank filled from a county source.

6 · Food: keep 14 days of food that needs no cooking and no fridge

PARTIALLY CONFIRMED FROM ACTION 6

BY WHEN	COST TO YOU	PROTECTS
15 Dec 2026	About $150–250, with a cooler (estimate)	Your household's meals

* Suggested formulation: check with a health worker before sharing.

Food reaches North Kohala by barge to Kawaihae, then by truck on Route 270 or 250, so the weak point is a closed road or a power cut, not empty shelves. USDA's outage rule: a closed fridge keeps food safe about 4 hours, a full freezer about 48 (24 if half full); throw out food above 40°F for more than 2 hours.

7 · Smoke and vog: keep fitted N95 masks, and set up one closed room with a HEPA air cleaner for anyone older or with heart or lung disease

PARTIALLY CONFIRMED | FROM ACTION 1 AND DO NOT, CARD 4

BY WHEN	COST TO YOU	PROTECTS
31 Dec 2026	N95 about $1–2 each; HEPA cleaner about $100–300 (estimates)	Lungs and hearts

The state health department says cloth masks will not protect you from ash; use a fitted N95. Masks, N95s included, do not stop vog's sulphur dioxide gas: for vog, stay indoors, limit exertion, and keep medicines to hand. The air cleaner stops when the power does, so pair it with step 3 if someone depends on it.*

8 · Fuel around the house: clear dry grass and debris within 5 feet of the house, and thin it out to 30 feet, before January

PARTIALLY CONFIRMED | FROM ACTION 5

BY WHEN	COST TO YOU	PROTECTS
31 Jan 2027	$0 with your own labour	The house

The 0–5 and 5–30 foot zones are standard Firewise guidance. Clear branches Lala left before the dry months cure them. The Hawai ʻ i Wildfire Management Organization's free home assessments are paused on Hawai ʻ i Island, so use its Firewise checklist (hwmo.org) and ask when assessments reopen.

9 · Health and money: flu and COVID shots, a spare week of prescriptions, a check on your insurance renewal, and, if you graze land, a rainfall-cover quote by 1 December

PARTIALLY CONFIRMED | FROM ACTIONS 7 AND 8

* Suggested formulation: check with a health worker before sharing.

BY WHEN	COST TO YOU	PROTECTS
15 Nov 2026; grazing cover by 1 Dec	Usually $0 if insured; premiums vary	Health, the house, the herd

Hawai ʻi follows the West Coast Health Alliance's 2026–27 advice: flu shots for all adults, COVID shots for everyone 65 and over and adults with risk factors, and a once-only RSV dose at 75 and over, or 50–74 with risk factors. Standard homeowner policies cover wildfire; the risk is non-renewal, and the state's last-resort insurer (HPIA) covers lava zones and condos, not North Kohala houses, so check your renewal date now. If you get SNAP and are 18–64, ask DHS whether you are exempt from the work rule; its clock has run since November 2025. If you graze land, ask a crop-insurance agent about USDA's rainfall-index pasture cover (action 8).*

What to ask of the people in charge

- Ask the **Department of Water Supply** whether every North Kohala well and booster has a generator hookup and a booked rental, and how long water lasts in an outage (action 2).
- Ask **Hawai ʻi County Civil Defense** when the plan for a Route 270 closure will be written and rehearsed (action 3).
- Ask the **West Kohala Wildfire Alliance** whether Hāwī and Kapa ʻau are inside its grant work, and how the town core can join a Firewise application (action 5).

WHAT THIS PLACE SHOULD DO — RANKED

Ranked by expected lives saved per unit of effort before the peak. Backing lenses are shown per action; an action several lenses reached

* Suggested formulation: check with a health worker before sharing.

independently is the stronger bet.

1. Build an opt-in register of North Kohala residents on powered medical equipment, the frail, and those with no car, and give each a backup-power and transport plan sized for a multi-day outage, by 30 November

PARTIALLY CONFIRMED | PUBLIC HEALTH · EMERGENCY MANAGER · FRONTLINE · PHILANTHROPIST · ECONOMIST | ADAPTATION INVENTORY: NONE FOUND

OWNER	BY WHEN	COST	PREVENTS
Hawai ʻi DOH, Hawai ʻi District, with Civil Defense and Kohala Hospital	30 Nov 2026	About $57,200: 44 batteries with solar panels at about $1,300 (estimate)	Deaths at home when power or roads are cut

The federal HHS emPOWER map counts 44 Medicare beneficiaries on electricity-dependent equipment in the district: 26 in 96755 and 18 in 96719. In the 2003 New York blackout, non-accidental deaths rose 25% (95% CI 12–41%; Anderson & Bell, *Epidemiology* 2012). In California, severe planned shutoffs raised same-day respiratory visits by 6%, and by 16% when smoke was present, with COPD worst affected (Wilner et al., *PLOS Medicine*, 27 October 2026). Kohala Hospital, a 25-bed critical-access hospital with a 24-hour emergency room, is the shelter and referral point. Hawaiian Electric's proposed $1,000 battery rebate is still awaiting the regulator and would pay about six months after approval, so it will not arrive before this winter.

Transferability: the mortality evidence comes from New York and California, not rural Hawai ʻi. Kapa ʻau and Hāwī are probably outside the planned-shutoff zone; the outages here come from wind and trees (about 2,100 North Kohala customers in November 2025; days without power after Lala). The emPOWER count is Medicare only, so it undercounts younger users.

2. Fit generator hookups and pre-book rental generators for the North Kohala wells and boosters that lack them, agree restoration priority with Hawaiian Electric, and sign standby water-hauling contracts, by 15 December

PARTIALLY CONFIRMED | HISTORIAN · EMERGENCY MANAGER · SKEPTIC · PUBLIC HEALTH · ECONOMIST | ADAPTATION INVENTORY: NONE FOUND

OWNER	BY WHEN	COST	PREVENTS
County Department of Water Supply, with Hawaiian Electric	15 Dec 2026	Transfer switches and rentals; not priced	Dry taps during a fire or outage

* Suggested formulation: check with a health worker before sharing.

Every North Kohala tap-water emergency on record began with a broken well or lost power, not drought: one of three Hāwī wells working in August 2025 (a 25% cut, then essential needs only from 1 September, with water hauled in, lifted 13 November); an outage on 13 November 2025 that put Hāwī #2 on a generator; and the 14 March 2026 storm. Part of this is done: DWS keeps a generator at one North Kohala well and at two boosters, because the line is a dead end, but countywide it has 18 generators for about 70 wells, and its generator plan is "informal." DWS said in 2024 that generators "CANNOT fully replicate" the grid.

Transferability: direct. A permanent generator at every well by December is not feasible (recent units cost $400,000–600,000 each and come from federal hazard grants), so this action is hookups, rentals, and contracts.

3. Write and rehearse a North Kohala evacuation plan for a Route 270 closure, with one-way outbound control on Route 250, an emergency lane kept open, and a rule that no exit closes while it is the last one, by 15 December

PARTIALLY CONFIRMED | EMERGENCY MANAGER · SKEPTIC | ADAPTATION INVENTORY: NOT CHECKED

OWNER	BY WHEN	COST	PREVENTS
Hawai ʻi County Civil Defense, with Hawai ʻi Police and HDOT	15 Dec 2026 (tabletop exercise)	Staff time	People trapped on one closed road

On 8 August 2023 fire closed Route 270 between mile markers 6 and 17 and forced the evacuation of Kohala Ranch, Kohala by the Sea, Kohala Estates, and Kohala Waterfront; FEMA's grant request put about 200 homes under threat. Upper Kohala Ranch left by Kohala Mountain Road, and Civil Defense has routed detours through the Ranch before; the plan should write that route down. No published North Kohala evacuation plan was found. The Lahaina inquiry found Maui police had no evacuation plan for the town, and its top recommendations call for standard plans for high-risk areas.

* Suggested formulation: check with a health worker before sharing.

Transferability: strong. Route 250 is a two-lane mountain road that has itself closed for floods and for Lala, so the plan must cover that case and keep a lane for fire engines.

4. Pair every kūpuna living alone, catchment household, and medical-device user with a named neighbour who checks on them in any outage, water notice, or fire, and sign them up for alerts at the door, by 1 December

PARTIALLY CONFIRMED | FRONTLINE · SKEPTIC · PUBLIC HEALTH | ADAPTATION INVENTORY: NOT CHECKED

OWNER	BY WHEN	COST	PREVENTS
Civil Defense, through the North Kohala Community Resource Center, churches, and county senior programmes	1 Dec 2026	Volunteer time; CERT training is free	Warnings and water notices that never reach people

The county's water notices go to county customers; catchment households hear nothing. Everbridge alerts reach only people who signed up. Volunteers set phones up, note who has no car (feeding action 1), and post water and shutoff notices in plain English and Ilokano at the post office, churches, and the Hāwī farmers market. Recruit block captains through a Civil Defense CERT course in Kapa ʻau (one was scheduled at the Old Kohala Courthouse in July 2024; CERT@hawaiicounty.gov). Kohala by the Sea was Hawai ʻi's first Firewise community, in 2004, so the habit of organising exists.

* Suggested formulation: check with a health worker before sharing.

Transferability: strong. The alert system exists; the missing piece is a person at the door. The size of North Kohala's Ilokano-speaking population was not found.

5. Clear roadside fuel on Route 270 and extend it to the Hāwī–Kapaʻau edge, remove Lala storm debris, and front-load the West Kohala Wildfire Alliance's grant work into November–January, before the grass cures

PARTIALLY CONFIRMED | EMERGENCY MANAGER · HISTORIAN · FRONTLINE · ECONOMIST · PHILANTHROPIST | ADAPTATION INVENTORY: NONE FOUND

OWNER	BY WHEN	COST	PREVENTS
Hawai ʻi Fire Department, with HDOT and the West Kohala Wildfire Alliance	31 Jan 2027	$300,000 already granted; debris at 75% FEMA share	Fire reaching homes in the 2027 dry season

On 29 July 2026 the state's HUI wildfire grants paid $1.8 million to eight groups, including $300,000 to the West Kohala Wildfire Alliance for manual and mechanical clearing and managed grazing in six Firewise communities. Debris on public land can be claimed under the Lala major-disaster declaration (8 October 2026, public assistance). No Firewise status was found for the Hāwī–Kapa ʻau core, and whether the Alliance's six communities include it could not be confirmed.

Transferability: direct. The panel disagreed on timing (see below): clearing now removes this year's growth before it cures, while the fire peak is likely spring and summer 2027.

6. Stock two weeks of no-cook food for about 500 North Kohala households and sign the outage playbook used after Lala, by 15 December

PARTIALLY CONFIRMED | FOOD SUPPLY CHAIN · FRONTLINE · PHILANTHROPIST | ADAPTATION INVENTORY: NONE FOUND

OWNER	BY WHEN	COST	PREVENTS
Kohala Food Hub, with St. Augustine's Episcopal Church and The Food Basket	15 Dec 2026	About $25,000–40,000 (estimate, unquoted)	Kūpuna without food when roads or power are cut

* Suggested formulation: check with a health worker before sharing.

After Lala, the Kohala Food Hub ran a hot-food distribution at St. Augustine's and handed out two thousand pounds of ice from the Mauna Kea Resort; The Food Basket moved nearly a million pounds island-wide. The playbook agrees who opens, who stocks, and where the ice comes from before the next outage. North Kohala will not run out of food: the barges keep coming. It runs out of cold, and of the means to drive to Waimea.

Transferability: direct. In-kind food only beats cash while stores and card readers are down; once they reopen, vouchers do more, as The Food Basket's $300,000 of grocery vouchers after Lala did. Winter wind outages (28,300 customers on 13 March 2026) justify a December date even though hurricane season is over.

7. Request USDA's good-faith SNAP work-rule exemption for Hawai‘i now, or publish its status, and screen North Kohala SNAP adults aged 18–64 for exemptions at once

PARTIALLY CONFIRMED | FOOD SUPPLY CHAIN | ADAPTATION INVENTORY: NONE FOUND

OWNER	BY WHEN	COST	PREVENTS
Hawai ʻi Department of Human Services	30 Nov 2026	Staff time	Benefits ending in the middle of the dry season

The 2025 federal budget law lets USDA exempt individuals in Alaska and Hawai ʻi from the new work rules through 31 December 2028 if the state asks and shows good faith (CRS R48552). The rule took effect in Hawai ʻi on 1 November 2025, so three-month clocks have been running for a year. Whether DHS has asked could not be confirmed; the 30 September 2026 date reported earlier was a penalty hold-harmless, not this exemption. About 43% of Hawai ʻi Island residents were food insecure in Hawai ʻi Foodbank's 2024–25 survey.

* Suggested formulation: check with a health worker before sharing.

Transferability: direct. This protects household cash, which matters more here than food prices: USDA forecasts food-at-home prices up 2.3% in 2026 and no El Niño price peak.

8. Ranchers buy USDA's rainfall-index pasture insurance before sales close (about 1 December) for January–April 2027, and file 2026 livestock forage claims by 1 March 2027

PARTIALLY CONFIRMED | ECONOMIST · FOOD SUPPLY CHAIN · HISTORIAN | ADAPTATION INVENTORY: NONE FOUND

OWNER	BY WHEN	COST	PREVENTS
USDA Risk Management Agency and Farm Service Agency (Hilo, 808-933-8334)	About 1 Dec 2026; LFP by 1 Mar 2027	Subsidised premium, quote-dependent	Forced sale of herds and ranch income lost

The Pasture, Rangeland, and Forage pilot is the one pre-arranged, triggered instrument found for this place. It is sold on Hawai ʻ i Island only, on 5-km grids built from Hawai ʻ i Climate Data Portal rainfall, in two-month intervals, and pays when the grid's rain index falls below the chosen coverage level, with no loss adjuster. At launch it covered about 600,000 acres and $13 million of protection, roughly $20 an acre at full loss. The 1 December closing date is confirmed for crop year 2025; confirm 2027's with an agent. The 2026 Livestock Forage Disaster Program is already open for Hawai ʻ i County; a 2027 drought would fall under the 2027 program, triggered by eight weeks at D2 on the US Drought Monitor.

Ranked last because it protects livelihoods rather than lives. In 1998–99 Parker Ranch cut its herd from 50,000 toward 30,000 head; destocking is how El Niño reaches Kohala's ranches.

* Suggested formulation: check with a health worker before sharing.

WHO DOES WHAT

RACI CHART

R = Responsible (does the work) · **A** = Accountable (one per action; signs off and answers for it) · **C** = Consulted (asked before) · **I** = Informed (told after). Built from the owners named in the verified actions. No one here has been contacted or has agreed to a role; every row is a proposal.

Action	DOH	Civil Defense	Kohala Hospital	DWS	HECO	Police	HDOT	HFD	WKWA	NKCRC & volunteers	Food Hub	DHS	USDA
1. Register and backup power	**A/R**	**R**	**R**		**C**			**I**		**R**	**I**		
2. Wells and outages		**I**		**A/R**	**C**			**C**					
3. Route 270 evacuation plan		**A/R**	**C**			**R**	**R**	**C**		**I**			
4. Neighbour checks and alerts	**I**	**A**		**C**						**R**			
5. Fuel and storm debris		**I**					**R**	**A/R**	**R**	**C**			
6. Food stock and playbook		**C**								**C**	**A/R**	**I**	
7. SNAP exemption											**I**	**A/R**	**C**
8. Ranch cover and claims		**I**											**A/R**

What the chart shows: Civil Defense touches six rows but is accountable for only two, which is right for a coordinator. No single agency carries the plan. The two rows that need a decision rather than money are 2 (whether DWS formalises its generator plan with Hawaiian Electric) and 7 (whether DHS asks for the exemption). The Food Basket supports row 6 alongside the Food Hub.

WHERE MONEY DOES THE MOST

FUNDING

The Philanthropist lens, reconciled against the ranked actions and verified. Placements are ranked by marginal lives saved per dollar. The envelope is bounded by what this place can spend before the peak, not by what it needs.

FLOOR	TARGET	CEILING	LARGEST UNCERTAINTY
$94,270	$144,270	$475,620	How many people under 65 depend on powered equipment (emPOWER counts Medicare only)

Floor: 44 batteries with solar panels at about $1,300 ($57,200) + 150 HEPA cleaners with a spare filter at about $190 ($28,500) = $85,700, plus a 10% fiscal-sponsor fee placeholder ($8,570) = $94,270. Target: floor + Kohala Food Hub outage reserve ($25,000) + a $25,000 match for a town-core HUI wildfire application = $144,270. Ceiling: 44 batteries ($57,200) + 300 cleaners ($57,000) + 10% fee ($11,420) + a $50,000 food reserve + a $300,000 HUI-scale fuel project = $475,620. Unit prices are estimates, not quotes, and NKCRC's real fee rate is unpublished. The target fell 30% in verification, from about $206,000: a $100,000 fuel bridge grant through HWMO was dropped, because HWMO is the grantmaker, not a fiscal agent.

* Suggested formulation: check with a health worker before sharing.

RECIPIENT	USE	AMOUNT	MECHANISM	BASIS	VERDICT
North Kohala Community Resource Center (fiscal sponsor), recipients named by Kohala Hospital and home health	Multi-day backup power for the 44 powered-equipment users on the action 1 register	$57,200	Pre-positioned stock	No cost-per-death study; Anderson & Bell 2012 (New York blackout) and Wilner 2026 (California shutoffs). NKCRC has fiscally sponsored since 2002; 2024 expenses $823,000.	PARTIALLY CONFIRMED
North Kohala Community Resource Center, with Kohala Hospital	HEPA cleaners for households with someone 65+ with heart or lung disease (action 1)	$28,500	Pre-positioned stock	Fisk & Chan, *Indoor Air* 2017 (Southern California smoke model): portable cleaners pass a benefit-cost test only over several years of use, best in 65+ homes. Vog not modelled.	PARTIALLY CONFIRMED
Kohala Food Hub (legal entity to confirm)	Outage reserve of no-cook food and ice (action 6)	$25,000	Pre-positioned stock	No cost study. Delivered hot food and ice after Lala (HPR, 21 Aug 2026); spending scale not confirmed.	UNVERIFIED SCALE
A Hāwī–Kapaʻau community group, with HWMO's Firewise support	Match and planning for a town-core HUI wildfire application (action 5)	$25,000	Matching grant	HUI requires a 25% match. Federal fire grants average about 3:1 benefit-cost (NIBS), mainly property.	PARTIALLY CONFIRMED

The funder's one thing: do not pay for the county's wells; that is the utility's job and its federal hazard grants. Pay for the 44 people who die if those wells, and the grid behind them, stop for three days.

* Suggested formulation: check with a health worker before sharing.

Do not do this

CONTRAINDICATED · ADVICE THAT WOULD COST LIVES HERE

Advice that circulates for this hazard and would cost lives here.

Do not wait for a siren or a phone alert before leaving

CONTRAINDICATED CONFIRMED

On 8 August 2023 Lahaina's outdoor sirens were not sounded, and with cell towers and power down, many residents never got the phone alerts. Alerts depend on someone deciding to send them and on a grid that a fire or a wind storm takes out.

Instead: sign up for Everbridge, keep a battery weather radio, and leave at the first advisory, or when fire is reported near your road.

Do not boil or filter tap water under a fire-related "do not drink" order *

CONTRAINDICATED CONFIRMED

After the August 2023 fires, Maui's advisory for Upper Kula and Lahaina said boiling would not make the water safe: pipes that lost pressure can draw in benzene and other volatile chemicals, which boiling does not remove.

Instead: drink stored, sealed water and follow the advisory to the letter. Catchment users follow For you step 5.

Do not run a generator in a carport, garage, or lanai, or near a window *

CONTRAINDICATED CONFIRMED

Carbon monoxide poisoning follows outages: in a 2019 California power shutoff, eight people in Campo were taken to hospital after a generator was set too close to their home.

* Suggested formulation: check with a health worker before sharing.

Instead: run it at least 20 feet outside, away from doors, windows, and vents (CDC), with a battery CO alarm inside.

Do not rely on cloth or surgical masks against smoke and ash, or on any mask against vog*

CONTRAINDICATED CONFIRMED

Hawai ʻi DOH: cloth masks "will not protect you" from ash. And no mask, N95 included, filters vog's sulphur dioxide gas.

Instead: a fitted N95 against smoke and ash; for vog, stay indoors in a closed room and limit exertion.

Do not wait to see which road stays open

CONTRAINDICATED PARTIALLY CONFIRMED

In Lahaina, barricades on the highway pushed drivers onto Front Street as the fire arrived. North Kohala has two roads out, and in 2023 one of them closed for a day.

Instead: know both routes, leave before either is closed, and follow responders' directions. Officials: write the last-exit rule into action 3.

WHAT HAS HAPPENED HERE BEFORE — THE RECORD

YEAR	EVENT	TOLL	WHAT CHANGED AFTER
1998–2000 (El Niño, then La Niña)	Island-wide drought; county emergency January 1998, extended to Hāmākua and North Kohala in July 1999	No deaths found; Parker Ranch cut its herd from 50,000 toward 30,000 head	10% water cut for Makapala, Keokea, and Niuli ʻi, still in force in January 2000
Oct 2006 (weak El Niño)	Kīholo Bay earthquake breaks the Kohala Ditch	No deaths found	About $8 million repair; water back in 2008
2009–10 (El Niño)	Hawai ʻi's first D4 drought, reaching leeward North Kohala in March 2010	No deaths found	USDA disaster designation for Hawai ʻi County, December 2009

* Suggested formulation: check with a health worker before sharing.

YEAR	EVENT	TOLL	WHAT CHANGED AFTER
Feb 2016 (very strong El Niño)	Leeward drought; some spots under 10% of January rain; Ponoholo Ranch weighing destocking	No deaths found	Nothing structural found
2021 (neutral)	Rockslide cuts the Kohala Ditch	Farm water lost	$10 million allocated in 2023, still unreleased in 2025
Jul 2021 (neutral)	Mana Road fire, Waimea (South Kohala), over 40,000 acres	2 homes destroyed; no deaths	About $1.2 million FEMA reimbursement, 14 months later
Aug 2023 (El Niño developing)	Akoni Pule / Kohala Ranch fire, about 1,000 acres; Route 270 closed, four subdivisions evacuated	No homes lost; no deaths	Federal fire grant; HUI grant to the West Kohala Wildfire Alliance in 2026
Aug–Nov 2025 (neutral)	Hāwī well failures: 25% cut, then essential needs only, water hauled in	No deaths found	Generator at Hāwī #2; plan still "informal"
Mar 2026 (neutral)	Storm outage, about 19,000 customers island-wide	No deaths found	Essential-needs water order for North Kohala
Aug 2026 (strong El Niño)	Hurricane Lala passes the island's south; 70 mph gusts in Kohala; Kohala Mountain Road closed	One traffic death in Kaʻū; none in Kohala	Major disaster declaration, 8 October (public assistance only)

The recurrence nobody has noticed: every North Kohala tap-water crisis on record began with a broken well or lost power, never with a dry aquifer. El Niño's drought is the slow hazard here; the fast one is the windy day that brings fire weather, falling trees, and a dead pump together. A caution against overreading: the three most recent water crises fell in neutral years, so El Niño loads the dice here rather than deciding them. ENSO phases are from NOAA's ONI table.

* Suggested formulation: check with a health worker before sharing.

WHERE THE EXPERTS DISAGREE — UNRESOLVED

- **Act before the winter peak** versus **the fire window is spring and summer 2027**. The Skeptic argued the danger on leeward Hawai ʻ i follows the dry winter, so the brief's "before the peak" has the timing wrong. The briefing keeps December dates for people and plans, and runs fuel work to January before the grass cures. What would settle it: Hawai ʻ i Fire Department ignition data by month for North Kohala.
- **Fund generators at every well** versus **fund the people, not the utility**. The Historian and Emergency Manager put well power first; the Philanthropist argued county wells belong to the county's budget and federal grants. Verification split it: hookups and rentals for DWS (action 2), batteries for people from philanthropy (action 1).
- **Planned shutoffs are the outage threat** versus **wind and trees are**. Four lenses built on Hawaiian Electric's shutoff zone; the verifier found Kapa ʻ au and Hāwī are probably outside it, and that the district's outages have come from storms. The actions hold either way; the address check (For you step 3) settles it per house.

WHAT NOBODY COVERED — BLIND SPOT

No lens could count people in North Kohala who depend on powered equipment and are under 65, or who live off-grid on catchment with no phone signal. Both groups fall outside emPOWER, the utility's lists, and the county's water notices, and both are the people most exposed when power and water fail together. Second, no lens looked at heat inside homes without air conditioning in a dry, still El Niño winter; in this season it is a smaller risk than fire and outages, but not zero for the people on the action 1 register.

HOW TO READ THIS — METHOD

The panel is author-built. Eight simulated expert lenses were used to generate these recommendations. Where they agree, that is a strong hypothesis, not a consensus of the field. Every recommendation was then independently checked for evidence, transferability to this location, and feasibility before the peak; the verdict tags carry the result.

Mortality figures are resolved at Climate Impact Lab impact-region scale, roughly 300,000 people. They are not neighbourhood figures. The El Niño projection covers *heat only*; drought, flood, fire, smoke, and crop failure are additional to it and are carried here by the historical record rather than by a published number.

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- **Scope.** The request gave a street address in Kapa ʻau. A single address has no data of its own, so the briefing covers the North Kohala district: Kapa ʻau (ZIP 96755) and Hāwī (ZIP 96719). Climate Impact Lab impact region, as the report records it: a Hawai ʻi County region, not resolved. Mortality figures exist only at that scale (impact regions average roughly 300,000 people); sub-district detail is used for hazard exposure and action design, not for death counts.
- **Exposure.** Climate Impact Lab, *451,000 at Risk* (September 2026, pre-peer-review): ECMWF SEAS5 seasonal forecasts (51-member ensemble) applied to temperature–mortality relationships across 24,378 regions, against a 1996–2025 baseline, for June 2026 – February 2027. It covers heat only. It is never added to the Impact Map's long-run projections or quoted from the map's GDP-denominated mortality. Where heat is not the dominant hazard, the historical record carries the hazard instead of a borrowed number.
- **Expert panel.** Eight simulated expert lenses (emergency manager, public health scientist, frontline responder, skeptic, economist, historian, philanthropist, and food supply chain analyst) each researched this place independently and in parallel, from web sources, to a fixed brief. The Kihei, Maui briefing (19 October 2026) supplied the state-level evidence pack: the Lahaina lessons, the do-not-boil advisory, and the shutoff rebate. Each claim taken from it was re-checked.
- **Reconciliation.** Duplicate actions were merged and backing lenses recorded. Convergence was tested for shared bias (whether the evidence behind an agreed action came from a setting like this one) before it counted in an action's favour. Contradictions were kept, not resolved by dropping a side. Actions were ranked by expected lives saved per unit of effort before the peak, each was given a named payer and trigger or marked unfunded, and the funding placements were reconciled against the ranked list.
- **Verification.** Seven verifier agents checked every recommendation on three tests: evidence (the primary source and its effect size), transferability (whether that evidence came from somewhere like here), and feasibility (whether it can be done here before the peak). Financial actions were also checked for availability in this jurisdiction, a named payer, a trigger threshold, and whether they already exist; this confirmed the rainfall-index pasture cover and found the 2026 forage program already open. Food actions were checked for whether each indicator is published at a usable frequency; the Urban Hawai ʻi CPI covers Honolulu only and is bimonthly, so no price trigger is proposed. The funding envelope was checked for real recipients, published unit costs, and duplication, then re-summed. The historical record was checked event by event, with ENSO phases from NOAA's ONI table.
- **Who does what.** The RACI chart is built from the owners named in the verified actions, with one accountable party per action. It is a proposal: no organisation named in it has been contacted or has agreed to a role.
- **Verdicts.** *Confirmed*: evidence, transferability, and feasibility all hold. *Partially confirmed*: holds after the corrections shown. *Unverified*: could not be checked to a primary source. *Contraindicated*: would do harm or waste the window here.

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- **Adaptation Inventory.** Not usefully consulted. Verifiers who reached the inventory found it rendered empty (no country selected, 0 adaptations), and one reached only the submission page. "None found" in the tags means no precedent was returned, not that one was ruled out; "not checked" means the verifier could not reach the entries.
- **Reader.** No role was given, so the For you section is written for a resident. The steps were taken from the verified actions and from what the panel found about households here; steps not drawn from a ranked action were verified on the same three tests, judged for one household with its own money.
- **Health wording.** Passages where a wording error could hurt someone (generators and carbon monoxide, medical equipment in an outage, masks and smoke, vaccines, and drinking water after a fire) are marked "* Suggested formulation: check with a health worker before sharing." They were checked against published sources, not by a health professional. They are suggestions, not medical advice, and anyone passing them on should confirm them with a doctor, pharmacist, nurse, or community health worker first. The marker was added on 2 November 2026.
- **Rules.** No cost is derived by dividing a published total, and no local figure is invented; where data is thin, the report says so and shows the containing geography's number with its arithmetic. Every source is listed below with its verification status.
- **Limits.** The panel and verifiers are AI research agents working from web-available sources, not human experts, and no one in this place was interviewed. The impact region was not resolved. Hawaiian Electric's shutoff map did not render, so whether any North Kohala address is inside a shutoff zone is unconfirmed. The 2016 ranch-specific figures (rain at 3% of normal, a 35% herd cut) could not be found in a primary source and were left out. North Kohala's 65+ share, its Ilokano-speaking population, and the Kohala Food Hub's legal entity were not found. All unit prices are estimates.

References · Season: CPC ENSO diagnostic discussion, 8 Oct 2026 (verified) · CPC Hawaiʻi outlook, 15 Oct 2026 (verified) · US Drought Monitor, 27 Oct 2026 (verified) · Chu, ENSO and Hawaiʻi rainfall, University of Hawaiʻi (verified)
Fire and evacuation: HI-EMA, Kohala fire FMAG, 8 Aug 2023 (verified) · Big Island Video News, Kohala evacuations, 8 Aug 2023 (verified) · Kohala wildfire update, 9 Aug 2023 (verified) · Kohala Mountain Road flood closure, 2019 (verified) · HPR, AG Lahaina report, Sep 2024 (verified) · AG Phase 2, ten key actions (verified) · Civil Beat, Lahaina sirens (verified) · NBC, Lahaina roadblocks (verified) · Hawaiʻi Police, Everbridge alerts (verified) · NWS Honolulu, weather radio (verified) · CERT course, Kapaʻau, 2024 (verified) · HWMO Firewise (verified) · HWMO home assessments (paused) (verified) · DLNR, 2026 HUI wildfire grants (verified) · Environment Hawaiʻi, Firewise outcome 2023 (verified)
Power, health, and water: Hawaiian Electric, shutoffs (verified) · Medical Needs Communication Form (verified) · Hawaiian Electric, proposed backup-power rebate (verified; not approved) · HNN, South Kohala shutoff, Nov 2025 (verified) · HHS emPOWER, ZIP counts (verified) · Anderson & Bell 2012 (verified) · Wilner et al. 2026 (verified) · Kohala Hospital (verified) · Fisk & Chan 2017 (verified) · Hawaiʻi DOH, ash and masks

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(verified) · Hawaiʻi DOH, vog and masks, 2025 (verified) · CDC, generators (verified) · 10News, Campo CO poisoning (verified) · Maui unsafe water advisory, Aug 2023 (verified) · UH CTAHR, catchment and ash (verified) · Hawaiʻi DOH, rain catchment (verified) · Hawaiʻi DOH, 2026–27 vaccines (verified) · DWS, North Kohala restriction, Aug 2025 (verified) · DWS, essential needs only, Sep 2025 (verified) · DWS, restriction cancelled, Nov 2025 (verified) · Storm water restriction, Mar 2026 (verified) · DWS, 14-day water (verified) · Civil Beat, Lala and DWS generators (verified)

Record: Star-Bulletin, Aug 1999 (verified) · Civil Beat, Kohala Ditch, Mar 2024 (verified) · Star-Bulletin, D4 drought, Mar 2010 (verified) · Tribune-Herald, Feb 2016 (verified) · HI-EMA, Mana Road reimbursement (verified) · Civil Beat, Hurricane Lala (verified) · Star-Advertiser, Lala declaration (verified) · NOAA ONI table (verified)

Money and food: USDA RMA, PRF Hawaiʻi pilot (verified) · Big Island Now, PRF launch (verified) · USDA FSA, 2026 LFP (verified) · NIBS, fire mitigation (verified) · ProPublica, NKCRC filings (verified) · HPR, Kohala Food Hub after Lala (verified) · HNN, Food Basket after Lala (verified) · Star-Advertiser, March 2026 outages (verified) · CRS R48552, SNAP changes (verified) · USDA FNS, SNAP implementation (verified) · USDA ERS food price outlook, Oct 2026 (verified) · BLS, Urban Hawaiʻi CPI (verified) · USDA, food safety in outages (verified) · Act 296 (2025), HPIA (verified)

Exposure data · Climate Impact Lab, 451,000 at Risk · Impact Map · Adaptation Inventory

Generated 30 October 2026 for Kapaʻau and North Kohala, Hawaiʻi Island. El Niño Ready is volunteer-run and is not affiliated with Climate Week NYC, The Climate Group, or Columbia University.

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